Contact

www.linkedin.com/in/edwin-bermudez (LinkedIn)

Top Skills

TypeScript
Bash
Go (Programming Language)

Edwin Bermudez

Senior Software Engineer & Cheminformatics Specialist | Full-Stack, DevOps & MLOps | Modernizing R&D IT & Scientific Data Pipelines
United States

Summary

I have spent the last 12+ years living at the chaotic intersection of writing code, building reliable enterprise infrastructure, and translating complex science into production software. My career started in the deep end of enterprise full-stack engineering at major financial institutions, eventually evolving into cloud architecture, DevOps, and MLOps.Today I bridge the gap between deep life sciences research and enterprise-grade technology. Whether I’m architecting cheminformatics data pipelines, containerizing heavy AI/ML workloads for pharmaceutical giants, or orchestrating automated CI/CD and continuous training (CT) pipelines, I make sure sophisticated scientific models don't just sit in research notebooks, but actually survive the harsh realities of enterprise production without melting the cloud budget.Professionally, I’m the guy who translates complex data science, chemistry, and engineering needs into stable, automated, cloud-first systems. Personally, I measure my success by the number of automated pipelines I’ve built that allow me to sleep peacefully through the night without getting emergency alerts at 3:00 AM. If you want to talk about high-performance architecture, scientific software design, or why Kubernetes is both a blessing and a curse, let's connect.

Experience

Cognizant

Senior Software Engineer & Cheminformatics Specialist
July 2022 - Present (4 years 4 months)

I bridge the gap between complex life sciences domain challenges and enterprise-grade software architecture. Leveraging my dual background in computer science and computational chemistry, I design and build scalable, production-ready software systems, data pipelines, and cloud infrastructures for global pharmaceutical and biotech clients.

My work centers on transforming intricate scientific concepts—such as cheminformatics workflows, molecular data structures, and lab automation—into high-performance, maintainable software. By translating deep scientific requirements into clean, modern code, I help client teams modernize their R&D IT ecosystems, accelerate data discovery, and bridge cross-functional gaps between research scientists and software engineering units.

Aetna, a CVS Health Company

Senior Software Engineer

November 2018 - June 2022 (3 years 8 months)

Led the architecture and delivery of scalable healthcare applications, successfully integrating full-stack technologies with advanced data and AI capabilities to improve user experiences. Spearheaded engineering best practices, mentored junior developers, and collaborated closely with cross-functional stakeholders to transform complex healthcare workflows into secure, high-performance digital solutions.

Wells Fargo

7 years 4 months

DevOps/MLOps Engineer

May 2016 - October 2018 (2 years 6 months)

Transitioned to a specialized role focused on bridges between software development, system operations, and early-stage machine learning infrastructure. Designed and automated secure continuous integration and continuous deployment (CI/CD) pipelines to accelerate software delivery cycles across enterprise applications. Architected robust infrastructure-as-code (IaC) solutions and containerized environments to streamline the deployment and monitoring of analytical models and high-availability services. Partnered with data science teams to optimize resource allocation, automate model deployment pipelines, and ensure data engineering workflows met stringent institutional standards.

Full Stack Engineer

July 2011 - May 2016 (4 years 11 months)

Designed, developed, and maintained robust end-to-end web applications, ensuring high availability and seamless integration between frontend user interfaces and backend database architectures. Collaborated with product teams to build secure, scalable financial services software while optimizing system performance and modernizing legacy codebases.

I spent significant time building custom ETL processes and experimenting with early machine learning libraries to introduce predictive capabilities into existing product suites, moving beyond standard CRUD operations.

Ebix
Software Engineer Intern
May 2010 - August 2010 (4 months)

My first true immersion into professional software development. I assisted the engineering team with documentation, basic feature implementation, and unit testing. It was here that I learned that great software isn't just about syntax, but about solving real-world business problems.

Education

Hodges University
Master of Science - MS · (August 2011 - May 2013)

Hodges University
Bachelor of Science - BS · (August 2007 - May 2011)

Nova Southeastern University
Doctor of Philosophy · (2021)